

Randy Tang

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SUMMARY

Software engineer with 3+ years building **high-throughput, real-time data pipelines** processing tens of millions of daily transactions at a global bank. Deep **Python** and **SQL** proficiency with hands-on experience in **event-driven architectures**, message queues, and performance optimization. Active personal trader (crypto and prediction markets) with a track record of building **trader-facing market-data and alerting tools**.

TECHNICAL SKILLS

Languages: Python (advanced), SQL, TypeScript, Java (fundamentals) | **Data & Systems:** High-throughput pipelines, AMPS message queues (pub/sub), concurrency, Pandas, Postgres, SQLite, Docker
Markets: Exchange APIs (Binance, Hyperliquid, Kraken), market data (OI, funding, order flow), prediction markets | **Practices:** Performance optimization, real-time pipeline design, code review, Git

EXPERIENCE

Software Engineer — Bank of America

Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS message queues

- Designed and maintained **real-time data pipelines** processing **tens of millions of transactions daily** across multiple services with retries, error handling, and load balancing.
- Built an extensible **reference-data framework**: event-driven pipeline of services connected via **AMPS message queues**, querying multiple external APIs per trade event and handling failures gracefully.
- Implemented a **T+1 trade reconciliation service** validating trade-ID uniqueness across counterparties and issuing corrections at millions of transactions per day.
- Developed a parallelized **high-volume backreporting service**, automating a previously manual process with substantial speed and throughput gains.
- Built a transformer service (internal model → outgoing XML) and DTCC response parser to **reconcile transaction statuses** in real time.
- Optimized reliability, latency, and throughput of production services by orders of magnitude; wrote Python/Pandas tooling to **analyze 50-100GB datasets**.

Data Analyst (Contract) — Microsoft

Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Processed and annotated datasets used to train ML models; built Python pipelines to clean, transform, and generate reports on user-collected data for the data science team.

Engineering Intern — Northrop Grumman

Fall 2019, Summer 2020

NX, ANSYS

- Designed and released 50+ engineering drawings including flight and test components; performed structural analyses for large assemblies and fastener stacks.

PROJECTS

- Market Data Dashboard** (Grafana, Twilio, Postgres, Docker) — Ingests **real-time market data** (open interest, funding rates) from **Binance and Hyperliquid APIs**; Grafana visualization dashboard with automated SMS alerts on signal triggers.
- Polymarket Scanner** (Flask, Discord API, SQLite) — Scans Polymarket for **large trades and unusual activity**; real-time alerting pipeline delivering notifications to Discord.
- Benchmark Agency Platform** (Next.js, TypeScript, Supabase, Python) — Multi-tenant **financial-intelligence dashboard** (revenue trends, equipment P&L, AI-generated recommendations) for a consulting agency with **3+ paying clients**; includes an 8-stage lead-gen data pipeline (Google Places API + LLM scoring, 20 tests).

EDUCATION

Cornell University, College of Engineering — Ithaca, NY

Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- CUAIR (Cornell Unmanned Aircraft)** — autonomous plane systems: obstacle avoidance, object classification, airdrop; top team at AUVSI SUAS competition.